Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/05/2025
PI Name:	Mason, Christopher Ph.D.		
Project Title:	Spatiotemporal Mapping of the Impact of Spaceflight on the Heart and Brain		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	New York	**State:**	NY
Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	12/01/2021	**End Date:**	11/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Costes, Sylvain Ph.D. (NASA Ames Research Center) Galazka, Jonathan Ph.D. (NASA Ames Research Center) Giacomello, Stefania Ph.D. (Kungliga Tekniska Hogskolan)		
Grant/Contract No.:	80NSSC22K0254		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To prepare for future human exploration missions far from Earth, NASA's Space Biology Program is seeking to build a better understanding of the effects of spaceflight and zero gravity on the biology of microorganisms, plants, and animals in spacecraft, the International Space Station (ISS), and also in ground-based analog experiments. The National Research Council recommended that NASA undergo studies to elucidate the effects of short and long duration spaceflight on the biology of all three categories of organisms. Technological advances in next-generation sequencing (NGS), spatial transcriptomics, and proteomics (spatial omics), several of which we describe below, create an unprecedented opportunity for in-depth molecular studies applicable to the purposes of NASA's Space Biology Program. This provides scientists, engineers, and clinicians a more comprehensive view of the functional dynamics of organisms as they evolve and respond to unique or highly selective environments including the ISS. Spaceflight causes changes in cell signaling pathways that are better understood only by increasing the analysis resolution level. In this project, we will deploy new technologies, i.e. spatial transcriptomics, single-nucleus RNA-sequencing, multi-omic spatial mapping (human and microbial), and systems biology algorithms to discover new insights relevant to the impact of spaceflight on human health. These data and methods will shed light on the complex biosystem dynamics that spaceflight causes in humans. We will be able to clearly dissect the gene expression changes occurring at the single-cell level, analyze how these changes affect the cell-cell genetic and physical interactions, and begin the first-ever in vivo human-microbial interaction maps from spaceflight. To do so we will conduct rigorous and cutting-edge omics analysis using two complementary platforms (10x Genomics Visium and Nanostring's GeoMx) with six main rodent organs collected throughout several past spaceflight missions and their corresponding ground controls. Our integrated biology approach will allow us to understand physiological, anatomic, and molecular mechanisms of adaptation and response in animals to spaceflight. For our organism-wide study we will leverage the extensive amount of samples collected throughout several Rodent Research (RR) missions which are accessible through the Life Sciences Data Repository (LSDA). Several of these specimens have already been allocated for Dr. Mason through LSDA. Our study will represent the first-of-its-kind in space biology and will provide foundational discoveries that will allow us to understand not only how astronaut conditions can be improved during spaceflight, but also how the changes induced by spaceflight can be translated into modern medicine to improve human health on Earth. Moreover, we will apply several statistical and machine learning techniques in order to predict changes induced by spaceflight at the organism level for future long-term missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We will be able to clearly dissect the gene expression changes occurring at the single-cell level, analyze how these changes affect the cell-cell genetic and physical interactions, and begin the first-ever in vivo human-microbial interaction maps from spaceflight. To do so we will conduct rigorous and cutting-edge omics analysis using two complementary platforms (10x Genomics Visium and Nanostring's GeoMx) with six main rodent organs collected throughout several past spaceflight missions and their corresponding ground controls. Our integrated biology approach will allow us to understand physiological, anatomic, and molecular mechanisms of adaptation and response in animals to spaceflight.
	Our goal for sequencing libraries from the International Space Station (ISS)-terminal group was completed. In total, three eyes from the Live Animal Return (LAR) spaceflight group, three eyes from the LAR ground control group, three eyes from the ISS-T spaceflight group, and three eyes from the ISS-T ground control group are in the sequencing queue. As an example, one flight eye and one ground control eye was processed with the NanoString GeoMx platform in addition to the flight eye from the last progress report. We also processed the RNA for each tissue. This image was featured as a selected abstract in the 2023 American Society for Gravitational and Space Research (ASGSR) meeting, and the image also won an art award at the conference. We are now in the process of uploading all the remaining data to NASA's GeneLab and the NASA Open Science Data Repository (OSDR). Of note, this work led to one of our lab's scientists (Dr. Eliah Overbey) getting a faculty position at University of Austin in the fall of 2024 and another member of the lab (Dr. Jiwoon Park) getting a post-doctoral fellowship in the Church lab at Harvard. This will extend the space biology and space medicine work to a new lab, which will continue to collaborate with our team and to use these data. Also, these funds helped several projects and informed best practices in the large Inspiration4 / Japanese Space Agency (JAXA) / NASA package in Nature, published in June and July of 2024. Some of these papers include the following. [Ed. Note: For additional information, see Cumulative Bibliography.] 1. Sanders LM, et al. Biological research and self-driving labs in deep space supported by artificial intelligence. Nature machine intelligence. March 23 5, 208–209 (2023). 2. Scott RT, et al. Biomonitoring and precision health in deep space supported by artificial intelligence. Nature machine intelligence. Mar 23, 2023; 5, 196–207 (2023). 3. Overbey EG, Kim J, Tierney BT, Park J, Houerbi N, Lucaci AG, Garcia Medina S, Damle N, Najjar D, Grigorev K, Afshin EE, Ryon KA, Sienkiewicz K, Patras L, Klotz R, Ortiz V, MacKay M, Schweickart A, Chin CR, Sierra MA, Valenzuela MF, Dantas E, Nelson TM, Cekanaviciute E, Deards G, Foox J, Narayanan SA, Schmidt CM, Schmidt MA, Schmidt JC, Mullane S, Tigchelaar SS, Levitte S, Westover C, Bhattacharya C, Lucotti S, Wain Hirschberg J, Proszynski J, Burke M, Kleinman AS, Butler DJ, Loy C, Mzava O, Lenz J, Paul D, Mozsary C, Sanders LM, Taylor LE, Patel CO, Khan SA, Suhail Mohamad M, Byhaqui SGA, Aslam B, Gajadhar AS, Williamson L, Tandel P, Yang Q, Chu J, Benz RW, Siddiqui A, Hornburg D, Blease K, Moreno J, Boddicker A, Zhao J, Lajoie B, Scott RT, Gilbert RR, Lai Polo SH, Altomare A, Kruglyak S, Levy S, Ariyapala I, Beer J, Zhang B, Hudson BM, Rininger A, Church SE, Beheshti A, Church GM, Smith SM, Crucian BE, Zwart SR, Matei I, Lyden DC, Garrett-Bakelman F, Krumsiek J, Chen Q, Miller D, Shuga J, Williams S, Nemec C, Trudel G, Pelchat M, Laneuville O, De Vlaminck I, Gross S, Bolton KL, Bailey SM, Granstein R, Furman D, Melnick AM, Costes SV, Shirah B, Yu M, Menon AS, Mateus J, Meydan C, Mason CE. The Space Omics and Medical Atlas (SOMA) and international astronaut biobank. Nature. 2024 Aug;632(8027):1145-1154. 4. Mason CE, Green J, Adamopoulos KI, Afshin EE, Baechle JJ, Basner M, Bailey SM, Bielski L, Borg J, Borg J, Broddrick JT, Burke M, Caicedo A, Castañeda V, Chatterjee S, Chin CR, Church G, Costes SV, De Vlaminck I, Desai RI, Dhir R, Diaz JE, Etlin SM, Feinstein Z, Furman D, Garcia-Medina JS, Garrett-Bakelman F, Giacomello S, Gupta A, Hassanin A, Houerbi N, Irby I, Javorsky E, Jirak P, Jones CW, Kamal KY, Kangas BD, Karouia F, Kim J, Kim JH, Kleinman AS, Lam T, Lawler JM, Lee JA, Limoli CL, Lucaci A, MacKay M, McDonald JT, Melnick AM, Meydan C, Mieczkowski J, Muratani M, Najjar D, Othman MA,

Task Progress:

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This work was also featured on the cover of Nature in August 2024:

Overbey EG, Kim J, Tierney BT, Park J, Houerbi N, Lucaci AG, Garcia Medina S, Bailey SM, Granstein R, Furman D, Melnick AM, Costes SV, Shirah B, Yu M, Menon AS, Mateus J, Meydan C, Mason CE. "Cover of the journal NATURE for article, 'The Space Omics and Medical Atlas (SOMA) and international astronaut biobank.'" Nature. 2024 Aug;632(8027):1145-54. https://

This special issue featured much of the work of this grant and comparisons to human spaceflight data.

Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Sanders LM, Scott RT, Yang JH, Qutub AA, Garcia Martin H, Berrios DC, Hastings JJA, Rask J, Mackintosh G, Hoarfrost AL, Chalk S, Kalantari J, Khezeli K, Antonsen EL, Babdor J, Barker R, Baranzini SE, Beheshti A, Delgado-Aparicio GM, Glicksberg BS, Greene CS, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Mason CE, Matar M, Mias GI, Miller J, Myers JG, Nelson C, Oribello J, Park S-m, Parsons-Wingerter P, Prabhu RK, Reynolds RJ, Saravia-Butler A, Saria S, Sawyer A, Singh NK, Snyder M, Soboczenski F, Soman K, Theriot CA, Van Valen D, Venkateswaran K, Warren L, Worthey L, Zitnik M, Costes SV. "Biological research and self-driving labs in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:208-19. Review. https://doi.org/10.1038/s42256-023-00618-4 , Mar-2023
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